Beyond Conventional Convolutional Neural Networks: Kolmogorov Arnold Networks for Accurate Urine Culture Classification

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***Abstract*—Urinary tract infections are among the most common infections found worldwide, the timely detection of which is necessary for effective cure and treatment. Although urine sample is a paradigm, it is often labor intensive, which can delay treatment decisions. The study proposes the use of deep learning models for the recognition of colony forming units in the urine sample images. Kolmogorov-Arnold Networks,which uses trainable activation functions to enhance representation of features when compared with other multilayer perceptron models, are explored deeply in this approach. Convolutional neural networks are used as standard models, whereas the three KAN-based architectures—K2AN, KAN-C-Norm, and KAN-C-MLP—are designed specifically for bacterial growth detection. Experimental results of the proposed system reduces diagnostic time, minimizes human error and provides faster clinical decision-making, when implemented on a publicly available pathological image dataset, demonstrating higher accuracy than conventional approaches.**

***Index Terms*—urinary tract infection, medical image classification, deep learning, convolutional neural networks, Kolmogorov Arnold networks, computer aided diagnosis, bacterial colony detection**

I. Introduction

The urinary system plays a vital role in filtering blood and eliminating waste through urine. Under normal conditions, urine is sterile or contains very low microbial levels; however, increased pathogen presence can lead to urinary tract infections (UTIs) [1], [4]. UTIs are among the most common infectious diseases worldwide and impose significant healthcare burdens [1], [2]. They may be acquired in the community or associated with healthcare care and are most frequently caused by uropathogenic *Escherichia coli* [1]. Infections can progress from the bladder to the kidneys, if neglected and not diagnosed, causing serious problem [1].

It is important to obtain a timely diagnosis for effective treatment. Traditional methods of examining urine samples under the microscope often takes a lot of time and require technical expertise [4]. In contrast, conventional techniques, the laboratory and point-of-care technique [2], provide reliable results, introducing a minor delay in treatment [2], [4].

Artificial intelligence, particularly the deep learning-based automated recognition system, has enhanced medical diagnostics. Convolutional neural networks such as GoogLeNet [18],VGG [17] and ResNet [16] deliver reliable and strong results in image analysis. However, these models still rely on multilayer perceptron models in clarity and robustness [9].

Kolmogorov-Arnold Networks(KANs) have emerged as an alternative architecture that has replaced fixed activation functions of convolutional neural networks with learnable functions, providing more flexible modeling of complex data sets [8], [11]. In this work, a publicly available urine sample image dataset [5], [7] have been used to train and evaluate multiple deep learning model along with the proposed Kolmogorov-Arnold Networks based architectures. The proposed method evaluation mainly compares various convolutional neural networks such as GoogLeNet [18], Tiny VGG [17] and many other models with the designed Kolmogorov-Arnold Networks variants such as **K2AN**,**KAN-C-Norm** and **KAN-C-MLP**, formed many by combination of KAN model along with a convolutional model and multilayer perceptron model, to evaluate their performance in image classification tasks. Experimental results shows that the accuracy of the proposed models achieve approximately 7.84% higher than traditional models. This indicates that the proposed KAN models have improved feature recognition and learning ability. The recent studies highlight the effectiveness and robustness of KAN architectures in medical diagnostics applications [21].

II. Related Works

A. Existing Approaches

This section analysis and reviews prior research in automated urinary tract infection diagnosis, deep learning based image analysis, Kolmogorov-Arnold Networks(KANs) and interpretable models.

Automated UTI Diagnosis: Recent researches have explored various automated approaches for UTI diagnosis with the help of Artificial intelligence highlights the importance of computational tools in modern diagnostic tasks. The introduction of a clinical microscopy dataset for UTI detection and illustration of the computer vision models such as PatchU-Net, as observed by Liou et al. , can achieve higher accuracy with lower variability in manual examination [4]. The growing need for faster automated testing, as well as the two conventional techniques was reviewed by Santos et al. [2]. Earlier and precise diagnosis is necessary for effective and faster treatment, given by the clinical observations made by Pietropaolo, particularly during theincrease in antimicrobial resistance [3].

Deep Learning for Image Classification: Automated diagnostic systems have been improved with the help of deep learning, particularly the deep convolutional networks that improve image classification through hierarchical feature extraction. These models such as GoogLeNet [18], VGG [17] and ResNet [16] persist as standard in medical imaging providing reliability and scalability. In addition, pretrained networks adapt to domain-specific datasets, enhancing accuracy and training efficiency when filtered down to labeled data, with the help of transfer learning approaches [20].

Kolmogorov-Arnold Networks and Interpretable Learning: Kolmogorov-Arnold Networks(KANs), introduced by Liu and Tegmark [8] as edge-function based networks, have emerged as an alternate architecture mainly focused on interpretability and computational efficiency, i.e., learning mathematical structures from data present. Optimised KAN variants can match traditional convolutional neural networks at the same time providing enhanced results [9]. Further, applications in time-series classification [13] and biological signal analysis [14] explore and demonstrate their potency in modeling and evaluating complex nonlinear relationships. Initial vision research and studies suggest that KAN-based architectures can process high-dimensional visual data productively [12], while providing additional improvements, including residual activations and initialization strategies, improving convergence stability [11].

Hybrid and Vision-Based Diagnostic Models: Application of KAN-based models for automated UTI diagnosis reported higher accuracy than conventional neural network models, through recent work by Dutta et al. [21]. Curated imaging datasets [5], [7] has advanced research progress by allowing consistent evaluation. Meanwhile, attention based models improve feature representation and recognition by enabling models to focus selectively on relevant regions of the image [19].

Table I summarizes related work in UTI diagnostics and KAN-based architectures.

TABLE I
Summary of Related Works

Category	**Authors**	**Contribution**
UTI Diagnosis	Kumar *et al.* [1]	Reviewed epidemiology and clinical aspects of UTIs.
	Santos *et al.* [2]	Surveyed diagnostic methods including POC techniques.
	Liou *et al.* [4]	Developed microscopy dataset for automated UTI detection.
CNN Models	Szegedy *et al.* [18]	Proposed GoogLeNet for efficient deep learning.
	Simonyan *et al.* [17]	Demonstrated depth advantages via VGG networks.
	He *et al.* [16]	Introduced residual learning for deep architectures.
KAN Models	Liu and Tegmark [8]	Introduced interpretable Kolmogorov Arnold Networks.
	Shukla *et al.* [9]	Compared KAN and MLP representations.
	Dong *et al.* [13]	Validated robustness of KAN models.
	Mitra [11]	Improved KAN convergence and stability.
Datasets	DaSilva *et al.* [5]	Provided urine culture image dataset.
	Brugger *et al.* [7]	Automated bacterial colony counting.
Applications	Dutta *et al.* [21]	Applied KAN models to automated UTI diagnosis.
	Zhuang *et al.* [20]	Surveyed transfer learning for limited datasets.

B. Research Gaps

Although significant research exists on aspects of automated medical diagnosis such as deep convolutional neural networks, interpretable learning models and clinical image diagnosis, most research emphasize the component independently. Accuracy, interpretability and efficiency within a unified framework have been emphasized in limited work. To bridge the gap between these research, the proposed system aims to evaluate conventional models alongside KAN-based models under standard experimental setup. This is achieved by introducing a diagnostic framework that manages performance, transparency and computational efficiency.

III. Proposed Method

A. Urine Culture Image Dataset Collection

This study used a publicly accessible urine culture image dataset [5], [6]. To ensure uniform imaging conditions for computer vision analysis, images were captured with the help of a controlled setup using a **12-megapixel** smartphone camera mounted inside acrylic enclosure, uniform illumination and distance. Incubation of the Petri dishes was carried out at **35 °C** for 24 hours, and at times up to 48 hours to confirm growth of microbes. The samples were categorized as positive, negative, and uncertain. **Positive** samples are those with at least two colonies of similar composition, plates without any growth of microbes were labeled **Negative**, and Uncertain samples showed a single colony or mixed morphologies [7].

Fig. 1. An example from the urine culture dataset. Image in the first column(on the left) represent the Positive sample, that in the second column(middle) represent the Negative sample, and that in the third column(on the right) represent the Uncertain sample.

The publicly available dataset consists of 1,500 labeled images across three classes: Positive(498), Negative(500), and Uncertain(502). While the size of the dataset is modest compared to large-scale baselines, the balanced class distribution as well as the controlled process provide reliable and effective training and evaluation. In addition, they increase the robustness of learned feature classification and representation. An example from the urine culture dataset is shown in Fig. 1.

B. Multi-Layered Perceptrons

Multi-Layered Perceptrons (MLPs), widely applied machine learning models, are basic supervised neural networks containing, mainly, input, hidden, and output layers, forming the foundation of many modern deep learning models. Each neuron computes a weighted sum then processed by a nonlinear activation function. They are expressed by the Universal Approximation Theorem, that states that, given necessary and sufficient parametes, an MLP can approximate any continuous function [9].

Universal Approximation Theorem: A feedforward neural network with a single hidden layer can approximate any continuous function on a compact domain. Formally, for any continuous function $f \in C([0,1]^n)$ and any $\varepsilon > 0$, there exists an integer N, weights $w_i \in \mathbb{R}^n$, biases $b_i, \alpha_i \in \mathbb{R}$, and a nonlinear activation function $\sigma(\cdot)$ such that

$$\left| f(x) - \sum_{i=1}^{N} \alpha_i \, \sigma(w_i^\top x + b_i) \right| < \varepsilon \quad \text{for all } x \in [0,1]^n.$$

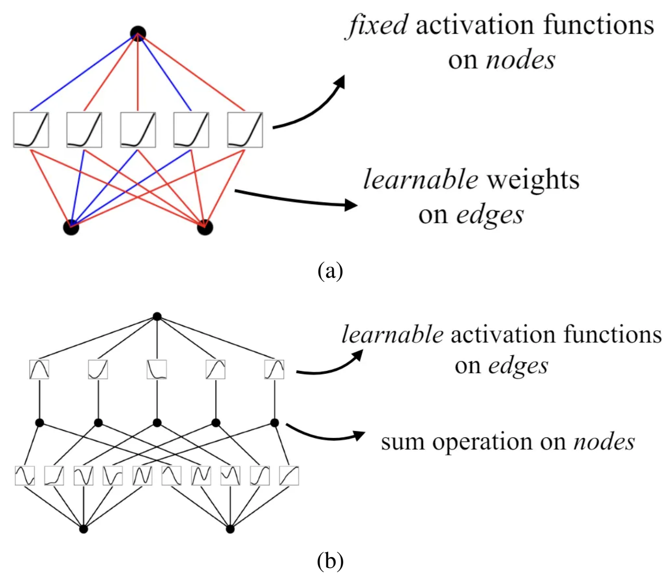


Fig. 2. Comparison of a Multi-Layered Perceptron (MLP) and a Kolmogorov–Arnold Network (KAN). (a) Multi-Layered Perceptron (MLP) Architecture (b) Kolmogorov–Arnold Network (KAN) Architecture

Figure 2(a) shows a MLP in which fixed activation functions operate at nodes while learnable weights lie on edges. Traditional MLPs often suffer from limited interpretability, high computational cost, and reduced accuracy for complex visual data [9], allowing the development of more advanced architectures.

MLP-Based Deep Architectures Considered: Image analysis can be achieved using several deep learning models derived from MLP principles. Some of the deep learning models include: **GoogLeNet** [18], which proposes Inception modules that perform parallel convolutions at multiple scales for effective feature extraction, having fewer parameters. Another approach is **VGG Networks** [17], which employ deep stacks of small convolution filters to learn hierarchical representations and morphology. **Vision Transformers**, typically requiring larger training datasets, uses attention mechanisms to capture long-range contextual dependencies [19]. In contrast, other CNN variants are optimized for better efficiency in limited resource environments [12]. While these models achieve strong performance in image classification tasks, they exhibit limited interpretability, motivating the exploration of various other approaches.

C. Kolmogorov–Arnold Networks

Kolmogorov-Arnold Networks(KANs), a recently developed neural model, was created to address the limitations of conventional multi-layered perceptrons, particularly the feature of their reliance on fixed activation functions. They were first introduced by Liu and Tegmark [8], to provide adaptable and interpretable representations by replacing the traditional parameterized activation function for weighted-learnable functions. **Kolmogorov–Arnold Representation Theorem**, which states that any continuous multivariate function can be can be represented as a finite composition of univariate functions and summation operators. This theorem forms the basis for this architecture.

There occurs a major difference in the way the nodes of Kolmogorov–Arnold Networks perform summation while edges learn activation functions, as compared to traditional networks, shown in Fig. 2(b) , supporting effective learning and improved interpretability [10], [12]. The activation functions are frequently parameterized using spline bases, specifically to achieve better convergence, stable gradients, and controlled nonlinearity [11]. Comparative research indicates that KAN-based models can perform on a variety of tasks as well as or better than traditional networks [9], [13], [14].

KAN-Based Structures: The examination of three KAN-based models were done to determine their effectiveness for the automated analysis of urine culture:-

- **K2AN:** employs stacked KAN-based convolutional layers for feature extraction, supporting dynamic nonlinear representation of colonies [9], [12].
- **KAN-C-Norm:** normalization steps are incorporated into KAN-based convolutional blocks to strengthen robustness against input variability, ensuring stable optimization [11], [12].
- **KAN-C-MLP:** a lightweight MLP classifier is combined with KAN-based feature extraction, providing balance in interpretability, flexibility, and effeciency [21].

These models display how function-based neural representations can improve medical image classification's efficiency, and robustness.

IV. Implementation and System Architecture

This section presents distinct training and deployment stages along with the implementation framework that allows end-to-end inference.

A. Integrated Deployment Framework

The trained K2AN, KAN-C-Norm, and KAN-C-MLP models are integrated by the framework to allow real-time interaction and classification of urine culture images. All models operate in inference mode to ensure prediction consistency. Before being passed to the models, the uploaded images undergo the same preprocessing steps used during training, testing, normalization, and format standardization. The output of the system displays scores for three categories—positive, negative, and uncertain—and assigns the final label based on highest score. This ensures consistency and efficiency between evaluation and deployment.

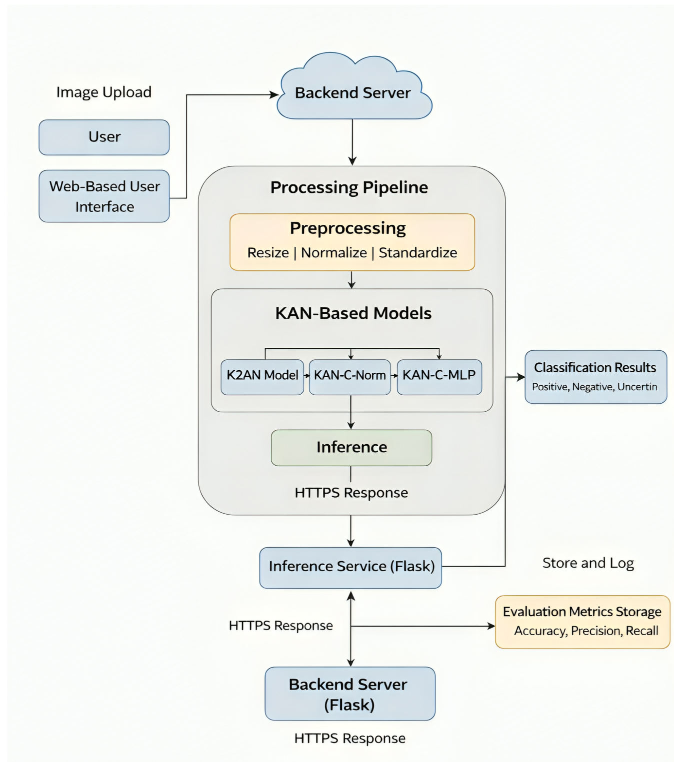


Fig. 3. System Architecture

B. Workflow of the System

An interactive web interface was implemented to show the real time working of the model. With the help of the backend models implemented, users can upload pictures from the urine dataset and analyse the classification result in real time. It only serves as a visualization layer and is used to show predictions clearly without going into technical details. Figure 3 shows how everything works together.

The first step here is to upload the image of urine sample through the web interface. The image is then sent to the backend for preprocessing using the trained KAN models along with the traditional deep learning models. The front end receives the predicted label, which is then displayed to the user.

Further, the deployment framework is only for testing and demonstration. With the help of this modular design, the integration of KAN-based diagnostic models into useful systems while maintaining a clear division between research and deployment can be visualized.

In general, the implementation highlights the usefulness of Kolmogorov–Arnold Network models for automated urine culture image analysis by bridging the gap between algorithmic development and practical applicability.

V. Experimental Results

This section assesses the suggested KAN-based models for urine culture image analysis and contrasts the results with traditional deep learning models.

A. Experimental Setup

A publicly available urine sample image dataset was used to carry out the experiment, which consisted mainly of three categories, that is, *Positive*, *Negative*, and *Uncertain*. The images in the dataset were standardized and configured to 224 × 224 pixels prior to training. 80% of the dataset was divided for training and the remaining for testing, allowing a fixed ratio to be applied across all models, which underwent training for 10 epochs and a batch size of 32, utilizing the cross-entropy loss for classification.

The experiments were executed within Visual Studio Code IDE, using PyTorch 2.0, on an NVIDIA T4 GPU configured with 16 GB RAM. Due to the recent emergence of KAN model, it was necessary to implement and integrate into the experimental framework.

B. Evaluation Metrics

The performance of the model was assessed utilizing standard classification metrics: *Accuracy*, *Precision*, *Recall*, and *F1-Score* [21]. In this context, TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively. The definitions of these metrics are as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

Training time (TrT) and inference time (IrT) were also recorded to evaluate computational efficiency.

C. Comparative Performance Analysis

A comparative analysis was performed between the proposed KAN architecture with traditional CNN and transfomer-based models such as VGG, GoogLeNet, and other variants. Maintainance of the same experimental conditions was a baseline for training and testing the models. To ensure fair and reliable comparison, results were reported as mean standard deviation over 10 independent runs. Table II depicts that the models have achieved strong baseline performance, with GoogLeNet and ResNet-18 showing stable results, while other variants of the model demostrates good feature learning but required more training time.

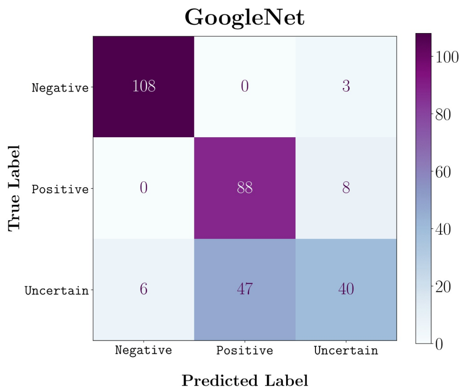
KAN-based models, in particular the **KAN-C-MLP** model achieved the highest overall performance, outperforming the baseline architectures across all evaluation metrics. The learnable activation function present in KAN-C-MLP model allows better modeling of visual pattern variations. Other variants of the KAN-based models also showed enhanced accuracy compared to traditional models. These KAN architectures also displayed lower variation across runs, indicating better robustness.

The confusion matrices in Fig. 4 demonstrate improved classification performance with fewer misclassifications. The experimental results highlight the effectiveness of KAN-based models for automated medical image classification.

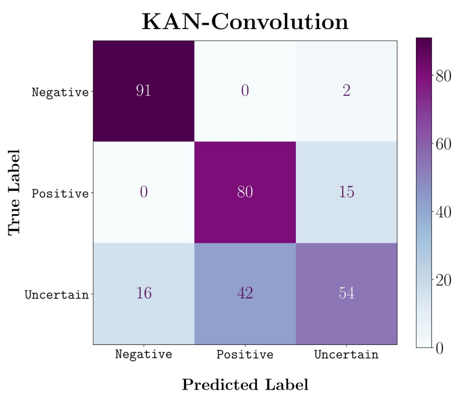
Although promising, the current framework classifies colony growth patterns rather than specific bacterial species. Future work will focus on integrating molecular diagnostic techniques and evaluating performance on larger and more diverse datasets to further assess generalizability.

TABLE II
Performance comparison of baseline and KAN-based models (mean with standard deviation).

Models	**Accuracy**	**Precision**	**Recall**	**F1 Score**	**TrT (s)**	**IrT (ms)**
Tiny VGG	67.66 (1.5391)	69.05 (0.9323)	65.48 (0.8235)	0.6722 (0.0862)	**512.4 (12.6)**	4.3 (0.5)
VGG-16	74.66 (2.7392)	77.37 (1.1347)	67.48 (1.0925)	0.7209 (0.0572)	923.1 (21.7)	6.8 (0.7)
VGG-19	73.33 (0.4823)	72.51 (1.9723)	72.33 (1.8241)	0.7242 (0.0214)	1058.6 (18.2)	7.4 (0.6)
GoogLeNet	78.66 (3.0419)	80.22 (1.8452)	74.25 (1.8091)	0.7712 (0.0627)	672.9 (14.5)	5.1 (0.4)
ResNet-18	78.01 (1.6528)	77.57 (0.7139)	77.05 (0.6482)	0.7731 (0.1012)	588.3 (16.9)	3.9 (0.3)
DenseNet	76.01 (1.5763)	74.74 (0.8945)	75.40 (0.7936)	0.7507 (0.1128)	892.7 (23.4)	5.9 (0.4)
CAiT	78.66 (2.6345)	78.57 (1.6345)	77.75 (1.8415)	0.7816 (0.0912)	1001.4 (27.1)	8.1 (0.6)
Xception	76.33 (1.8724)	75.34 (1.3497)	75.92 (1.1095)	0.7563 (0.0785)	1187.9 (22.5)	7.6 (0.5)
K^2AN	75.21 (1.1187)	75.55 (0.4921)	73.63 (0.2839)	0.7458 (0.0063)	678.5 (12.8)	4.7 (0.3)
KAN-C-Norm	86.95 (0.5634)	75.57 (0.4283)	74.02 (0.3991)	0.7479 (0.0127)	532.9 (11.6)	3.5 (0.3)
KAN-C-MLP	**87.16 (0.9654)**	**77.03 (1.2136)**	**68.01 (1.1906)**	**0.7224 (0.1921)**	529.3 (10.4)	**3.2 (0.2)**



(a)



(b)

Fig. 4. Confusion matrices of the top-performing models:
(a) GoogLeNet (78.66%); (b) KAN-C-MLP (proposed) (87.16%).

VI. Conclusion and Future Scope

Recent development in artificial intelligence have enhanced automated diagnostic system along with the accessibility of large medical datasets. In traditional urinary tract infection diagnosis, urine sample analysis is labor extensive process, requires technical expertise, and may introduce interpretation bias in efficient treatment. This research mainly investigates deep learning based image processing for automated classification of urine sample images to reduce time and improve accuracy.

Kolmogorov-Arnold Networks (KANs) and their variant models consistently achieve higher performance than conventional multilayer perceptron and convolutional neural network models, achieving standard accuracy of approximately 80%. This is clearly shown by the experimental results of the proposed system. As a proof of concept for real time deployment, a lightweight web interface was developed, allowing users to upload urine sample images and obtain predictions instantly.

Although the system's performance is promising, it currently classifies colony growth patterns but does not recognize specific species of bacteria, which may lead to classification error or class imbalance of various other pathogens. To address the limitations, future work will integrate image analysis with molecular methods such as expanding datasets, training, and evaluating multi-center performance metrics and include uncertainty estimation.

In conclusion, the proposed system implements and demonstrates that KAN-based models provide a promising solution for accurate, efficient, and automated classification and analysis of clinically viable urine culture.

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